

COMP202 Week 1 - solutions to exercises

Question 1. *John gave me the following argument, claiming to prove that $n^2 = O(n)$. What is the flaw in John's argument?*

Attempted proof by John

I'll show this by induction on n . Essentially, I'll show that n^2 is always at most a constant times n , which is what $n^2 = O(n)$ means.

- The **inductive hypothesis**: n^2 is at most a constant times n .
- The **base case** is clearly fine since the inductive hypothesis is true for $n = 1$.
- **Inductive step**: Assume the inductive hypothesis is true for $n = k \geq 1$. That is, $k^2 \leq ck$ for a constant c . I'm now going to show that the inductive hypothesis is also true for $k + 1$. That is, I'll show that $(k + 1)^2$ is also at most a constant times $k + 1$, which will prove what we want.

$$\begin{aligned}(k + 1)^2 &= k^2 + 2k + 1 && \text{expanding the square} \\ &\leq ck + 2k + 1 && \text{as we assumed hypothesis true for } n = k \\ &\leq ck + 3k && \text{since } 1 \leq k \\ &= (c + 3)k && \text{collecting common terms} \\ &\leq (c + 3)(k + 1). && \text{as } k \leq k + 1\end{aligned}$$

But since c was a constant, so is $c + 3$, so $(k + 1)^2$ is at most $(c + 3)$ times $k + 1$.

Thus, $n^2 = O(n)$.

Solution: Recall from the definition that $f(n) = O(g(n))$ means there are constants c and n_0 such that:

$$f(n) \leq c \cdot g(n) \text{ for all } n \geq n_0.$$

Crucially, note that c must be independent of n (i.e., **one cannot vary the value of the constant depending on n**), which is not the case in the attempted proof where $k^2 \leq ck$ but $(k+1)^2 \leq (c+3)(k+1)$: the value of ‘ c ’ for $k+1$ is larger than that for k . In other words, the induction hypothesis has not been presented formally enough: one must change it to something of the form “ $n^2 \leq cn$ for all $n \geq 2$ ”, but this is not true as observed above. $\square$

Question 2. $f(n)$ is said to be $o(g(n))$ if $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0$. Similarly, $f(n)$ is said to be $\omega(g(n))$ if $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \infty$ (or equivalently, if $g(n)$ is $o(f(n))$). Prove that $n! = \omega(2^n)$ and $n! = o(n^n)$.

Solution:

•

$$\frac{n!}{2^n} = \frac{n \cdot (n-1) \cdots 1}{2 \cdot 2 \cdots 2} = \frac{n}{2} \cdot \frac{n-1}{2} \cdots \frac{1}{2} \geq \frac{n}{4},$$

which goes to infinity as $n \rightarrow \infty$.

•

$$\frac{n!}{n^n} = \frac{n \cdot (n-1) \cdots 1}{n \cdot n \cdots n} = \frac{n}{n} \cdot \frac{n-1}{n} \cdots \frac{1}{n} \leq \frac{1}{n},$$

which goes to 0 as $n \rightarrow \infty$.

$\square$

Question 3. Put these functions in order so that if $f(n) \in O(g(n))$ (i.e. $f(n)$ is Big-Oh of $g(n)$), then $f(n)$ appears before $g(n)$ in the list. Group together functions that have the same asymptotic order of growth (i.e. $f(n) \in \Theta(g(n))$). In each case provide some justification to your claim.

$$\begin{aligned} & n^2, \ n \log n, \ n^3 + \log n, \ \sqrt{n}, \ n^2 + 2n \log n, \\ & \log \log n, \ 17 \log n, \ 10n^{\frac{3}{2}}, \ n^5 - n^4 + 2n, \\ & 5n^2 \log \log n, \ 3n^2 + n^3 \log n, \ n + 6 \log n \end{aligned}$$

Solution: Here is the list, where I have grouped together the functions having the same asymptotic order of growth (there's only one such pair):

$$\begin{aligned} &\log \log n, \ 17 \log n, \ \sqrt{n}, \ n + 6 \log n, \ n \log n, \\ &10n^{\frac{3}{2}}, \ \{n^2, n^2 + 2n \log n\}, \ 5n^2 \log \log n, \\ &n^3 + \log n, \ 3n^2 + n^3 \log n, \ n^5 - n^4 + 2n \end{aligned}$$

Essentially, most of these can be demonstrated using the fact that $\log n \leq n$ for all $n \geq 1$, and hence, $\log \log n \leq \log n$ for all $n \geq 2$.

Also, we know that $\log n \leq \sqrt{n}$ for all $n \geq 1$. If you know any calculus, this is easy to prove. First show that $\lim_{n \rightarrow \infty} \frac{\log n}{n} = 0$ using de L'Hôpital's Rule. This means that $\log n = o(n)$, and, as mentioned in class, this implies that $\log n = O(n)$. (I basically said that you can assume the fact that $\log n = O(n)$, and that $\log n = O(n^{\frac{1}{k}})$ for any fixed $k \geq 1$.)

We can easily make statements like $2n \log n \leq 2n^2 \leq 2n^2 \log \log n$ for large enough n . Similarly, $n^3 \log n \leq n^4 \leq n^5 - n^4$ for "large enough" n . These statements can be more formally proved.

Recall also that $n^{\frac{3}{2}} = n \cdot n^{\frac{1}{2}} = n \cdot \sqrt{n}$.

□

Question 4. *Prove by mathematical induction that Algorithm Minimum-Element from the introductory lectures is correct.*

Solution: We argue by induction that for any integer $j \geq 1$, the algorithm has in variable *min* the minimum in the sub-array $A[1..j]$. Induction base for $j = 1$ is obviously true due to line 1.

For the induction step, suppose for the induction hypothesis, that *min* has the current minimum in the subarray $A[1..j]$ and consider iteration $j + 1$ in the for loop. If in line 6 we have that $A[j + 1] < \text{min}$, then that means that the minimum of the sub-array $A[1..j + 1]$ is in fact $A[j + 1]$, which the algorithm correctly updates in line 7. Otherwise, the current value of *min* is also minimum in the sub-array $A[1..j + 1]$. Thus after $j + 1$ -st iteration, *min* has the current minimum in sub-array $A[1..j + 1]$. This finishes the inductive argument.

Now, using this fact for $j = \text{length}[A]$, we conclude that the algorithm returns as final *min* the correct minimum in the whole array. □